

Junior Makers · Class 26

# Smart Kitchen Scale

Weigh anything to the gram, and make it smart

Load cell

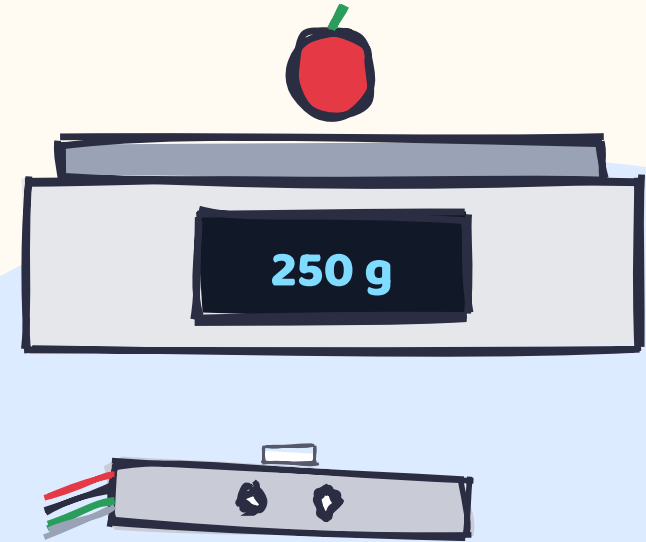
Calibration

OLED

Target alarm

Did you know?

Digital scales in shops must be checked by the government so customers always get the weight they pay for.





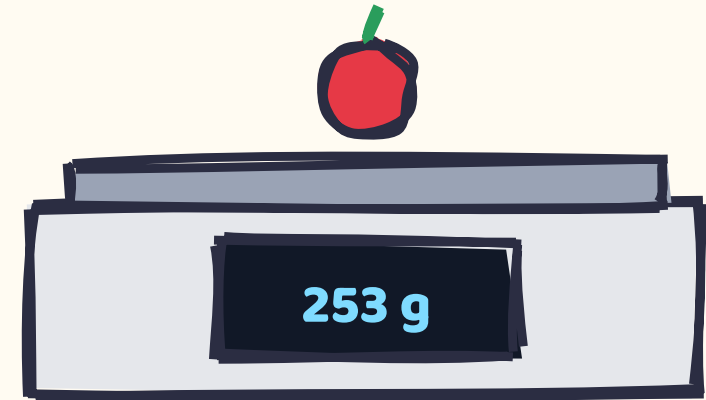
Let's think...

# How does a scale know the weight?



## No springs?

Old scales had a big spring that stretched, and a needle that moved. Digital scales have no needle and almost nothing moves. Yet they say **253 g** instantly. What is hiding inside?



A tiny spring?

Magic chip?

A bending metal bar!

**Today:** you build a **digital kitchen scale** with a load cell, then make it smart with a **target-weight alarm**.





## Learning objectives

# By the end of this class I can...



1

**Load cell:** explain how a **load cell** turns a tiny bend into a number**Proof:** explain 'strain' with the ruler example

2

**Calibrate:** tare (zero) and **calibrate** my scale with a known weight**Proof:** my scale reads a 100 g water cup within 5 g

3

**Display:** show the weight in grams on an **OLED**, with a tare button**Proof:** pressing the button sets it back to 0 g

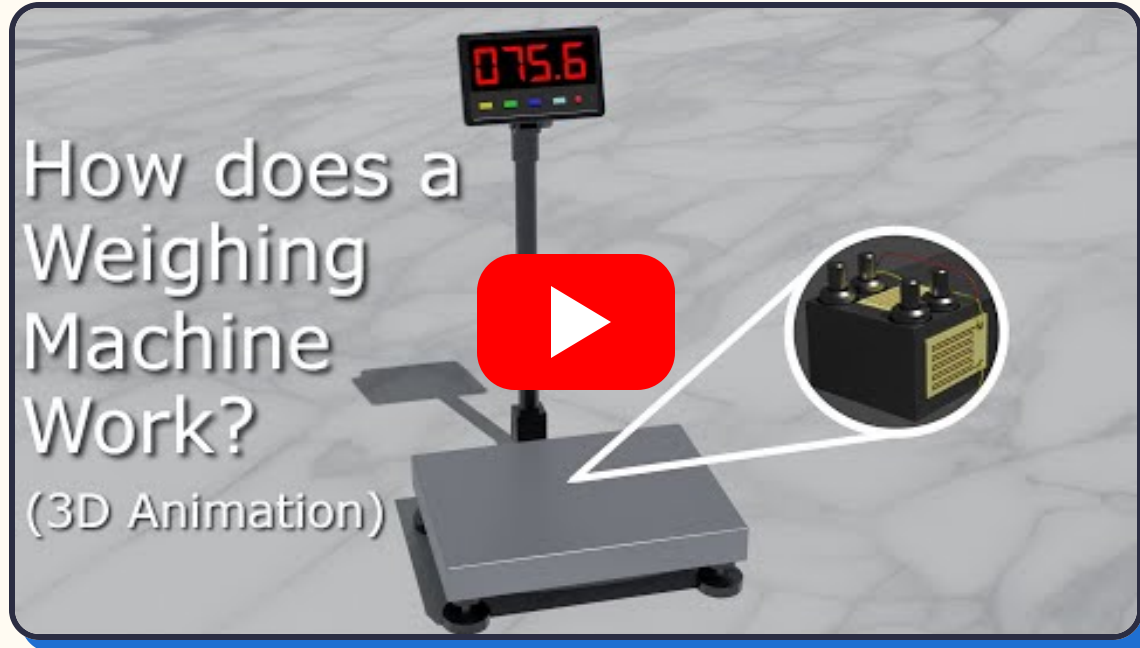
4

**Smart feature:** add a **target-weight alarm** for recipes or portions**Proof:** the buzzer beeps when I reach the target



Objective 1 • Load cell • video

# Video time!



## How does a Weighing Machine Work?

3D Requiem • about 4 minutes • [youtu.be/IZafPys6x5M](https://youtu.be/IZafPys6x5M)

### Watch for...

- 1 What part of the scale **bends** when you put something on it?
- 2 How does a tiny bend become a **number**?
- 3 Why must a scale be set to **zero** first?



**After the video:** Press gently on a ruler on the desk. Can you see it bend? A load cell bends far less than that!



## Objective 1 • Load cell

# A load cell is a bar that bends a tiny bit

## The idea

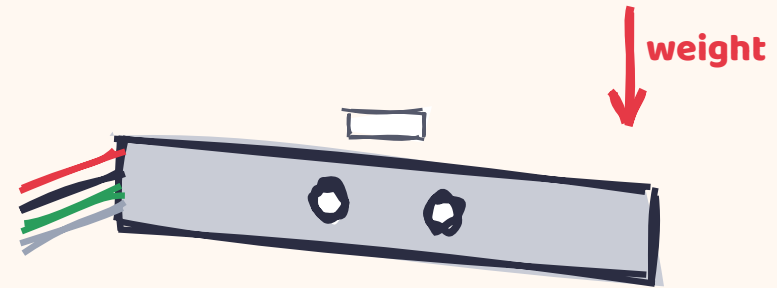
The load cell is an aluminium bar with a **strain gauge** glued on: a thin zig-zag wire. When weight bends the bar (much less than a hair's width!), the wire stretches and its **resistance changes** very slightly.

The change is so tiny that we need an **HX711** chip to amplify it and turn it into a number.

### Gentle!

Never press on the load cell with your fingers or drop things on it: too much force bends it for ever.

## Example: a ruler on a desk



Hang a ruler off the desk and press the end: it bends. Press harder: it bends more. The load cell bends the same way, just far, far less, and a sensor measures exactly how much.



Class game

# Guess the weight



## How to play

1. Pass around 5 mystery objects: a pencil case, an apple, a water bottle, a book, a shoe.
2. Each team writes its **guess** in grams for each one.
3. Keep the guesses secret until our scale is built.
4. At the end of class, weigh them all. Which team was closest overall?

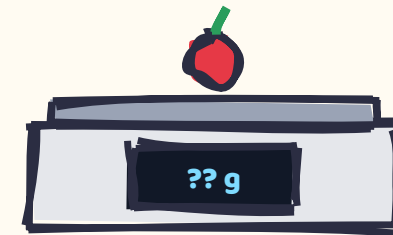
## Then discuss

Were your guesses too high or too low? Why are humans bad at guessing weight?



## Big idea

We need instruments because our senses are not precise. That's why we build and calibrate scales.





Build it

# Our scale kit (per group)

**Arduino Uno**

The scale's brain

**5 kg load cell + HX711**

Senses and amplifies the bend

**OLED display**

Shows the grams

**Push button + 10k**

The TARE (zero) button

**Buzzer + LED**

Target alarm

**Cardboard**

Base and weighing plate

The load cell must be fixed at ONE end to the base and hold the plate at the OTHER end, so it can bend. Use the screw holes.

**Max 5 kg**

Never put more than 5 kg on it. A heavy textbook pile could ruin it.



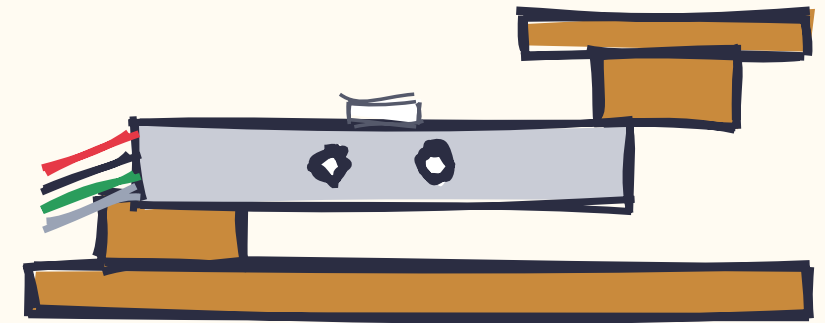


Objective 1 • Load cell • Build step 1 of 4

# Build and wire the scale



| Part                    | Connect to    |
|-------------------------|---------------|
| Load cell red / black   | HX711 E+ / E- |
| Load cell white / green | HX711 A- / A+ |
| HX711 VCC / GND         | 5V / GND      |
| HX711 DT / SCK          | pin 3 / pin 2 |
| OLED SDA / SCL          | A4 / A5       |
| Button (10k to GND)     | pin 4 and 5V  |

**fixed end • free end**

## The Z shape

Base → spacer → load cell → spacer → plate. The gaps let the bar bend.



Objective 1 • Load cell • Build step 2 of 4

# See the raw numbers

## Setup

HX711 init DT 3 SCK 2

## Loop

serial print HX711 read raw value

wait 500 ms

## Test

## Raw number

Empty plate

Press very gently

100 g of water in a cup

## Big, strange numbers

You'll see numbers like 84,512 or -12,300. They change when you add weight, but they are NOT grams yet. We need to **calibrate**.

## Wrong way?

If the number goes DOWN when you add weight, swap the white and green wires (or turn the cell over).



## Objective 2 · Calibrate

# Tare and calibrate

## The idea

**Tare:** with nothing on the plate, tell the scale 'this is 0 g'.  
(Bakers tare to ignore the bowl!)

**Calibrate:** put on something you KNOW the weight of. Water is perfect: **100 ml of water = 100 g**.

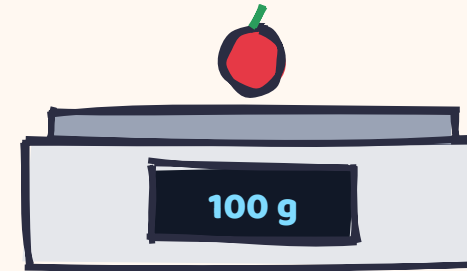
**scale factor = raw change ÷ known grams**

e.g. raw went up 21,400 for 100 g → factor = **214**

## Tip

Tare the empty cup first, then pour exactly 100 ml using a measuring jug or syringe.

## Example: a new ruler with no numbers



**100 ml water = 100 g**

Imagine a ruler with lines but no numbers. You measure a 10 cm card and write '10' at that line. Now the ruler is useful! Calibration writes the numbers on our scale.



Objective 3 • Display • Build step 3 of 4

# Grams on the screen

## Setup

HX711 init DT 3 SCK 2

HX711 set scale 214 (YOUR factor)

HX711 tare

## Loop

set grams to round( HX711 get weight )

if button pin 4 pressed then HX711 tare

OLED clear • text size 3 • print grams • print ' g' • show

wait 200 ms

| Test              | Should be | We got |
|-------------------|-----------|--------|
| 100 ml water      | 100 g     |        |
| 200 ml water      | 200 g     |        |
| Empty, after tare | 0 g       |        |

**Not quite?**

Reads 95 g for 100 g? New factor =  $214 \times 95 \div 100$ .  
Try again!



## Objective 2 • Calibrate

# Accurate and precise



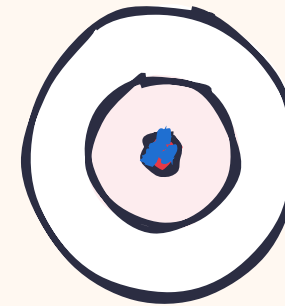
## The idea

**Accurate:** the reading is close to the true weight (100 g reads 100 g).

**Precise:** the reading is the same every time (100, 100, 101, 100).

Test it: weigh the same object **5 times**. Do the numbers jump around? Take the **average** of 10 readings to smooth out the wobble.

## Example: darts on a board



**both!**

Darts all bunched in a corner: precise, not accurate. Darts spread around the bullseye: accurate on average, not precise. All in the bullseye: both! Good scales are both.





## Objective 4 · Smart feature

# Make it smart: a target alarm



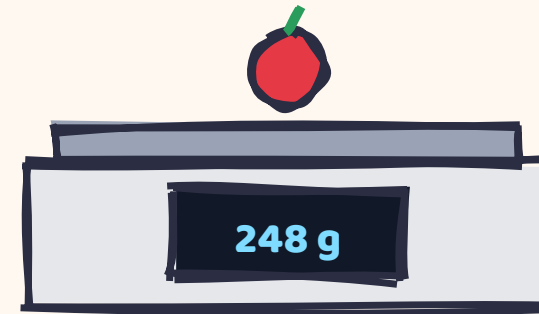
## The idea

A **target weight** turns a scale into a helper:

- Recipe mode: 'pour flour until **250 g**'
- The LED turns on at **90%** of the target (slow down!)
- The buzzer beeps at the target

Use **if** blocks to compare the weight with the target.

## Example: filling a glass



When you pour water into a glass, you slow down as it gets near the top. Our LED is the 'slow down' warning and the buzzer is the 'stop!'.





Objective 4 · Smart feature · Build step 4 of 4

# Recipe alarm

**Setup: set target to 250****Loop**

(read grams and show on OLED as before)

if grams  $\geq$  target  $\times$  0.9 then LED pin 12 ON else LED OFFif grams  $\geq$  target and not beeped then

tone pin 11 1000 Hz 300 ms · set beeped to true

if grams &lt; 5 then set beeped to false

## Why 'beeped'?

Without it, the buzzer would beep forever while the bowl sits on the scale. It resets when the plate is empty.



## Change target

Use the button: short press = tare, long press = add 50 g to the target.





Real world

# Load cells everywhere



## **Shops**

Fruit and meat scales that print a price



## **Airports**

Check-in belts weigh your luggage



## **Weighbridges**

Weigh whole trucks on the motorway



## **Hospitals**

Baby scales and beds that weigh patients



Why must a shop's scale be checked by inspectors?

## **Fairness**

If a scale reads 10% heavy, every customer pays 10% too much. Calibration is about honesty.





Objective 4 • Smart feature • test it

# Guess the weight: results!



| Object       | Our guess | Scale says | Difference |
|--------------|-----------|------------|------------|
| Pencil case  |           |            |            |
| Apple        |           |            |            |
| Water bottle |           |            |            |
| Book         |           |            |            |
| Shoe         |           |            |            |

## Total difference

Add up all the differences (ignore minus signs). Lowest total wins!



Test like an engineer

# Bug hunt



| Bug                          | Likely cause                   | Did it happen? |
|------------------------------|--------------------------------|----------------|
| Reading always 0 or stuck    | DT/SCK swapped or loose wire   |                |
| Number goes down with weight | A+/A- swapped                  |                |
| Reading slowly drifts        | load cell warming up / bumping |                |
| Wrong grams                  | scale factor not calibrated    |                |
| Very jumpy numbers           | plate touching the base        |                |

## Debug checklist

Check the Z-shape: the plate must ONLY touch the load cell, nothing else.



Let it sit for 1 minute after switching on, then tare. Electronics like to warm up.





Level up!

# Scale challenges



## Choose your challenge



**Price scale:** type a price per kg; show the cost of what's on the plate.



**Food waste:** weigh lunch leftovers each day and add up the week.



**Counting scale:** weigh 10 coins, then COUNT a pile of coins by weight!



**Units:** a button that switches between grams and ounces.

## Think like an engineer

If 10 identical coins weigh 52 g, how many coins are in a pile that weighs 260 g?



## Boss level

Make the scale average 10 readings for a steadier display.





## Objective check

# Can I really do it? Prove it.



**01** What bends inside the scale, and what does the HX711 do?



**02** Explain tare and calibrate. What is YOUR scale factor?



**03** Weigh 100 ml of water. How close is your scale?



**04** Show your recipe alarm reaching the target.



## Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

**Teacher:** Scale accuracy contest: weigh the teacher's mystery object. Closest scale wins.





Show what you know

# What did we learn?

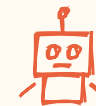
- ☒ A load cell is a bar that bends a tiny bit under weight
- ☒ The HX711 amplifies the tiny change into a number
- ☒ Tare = set zero; calibrate = match a known weight
- ☒ Accurate = close to the truth; precise = repeatable
- ☒ if blocks turn a scale into a smart helper



**Exit ticket:** your scale reads 90 g for 100 ml of water. What do you change?

## At home

Weigh 3 things at home on a kitchen scale. Bring the numbers to test our scales!



**Next class:**  
Arcade Claw  
Machine!

